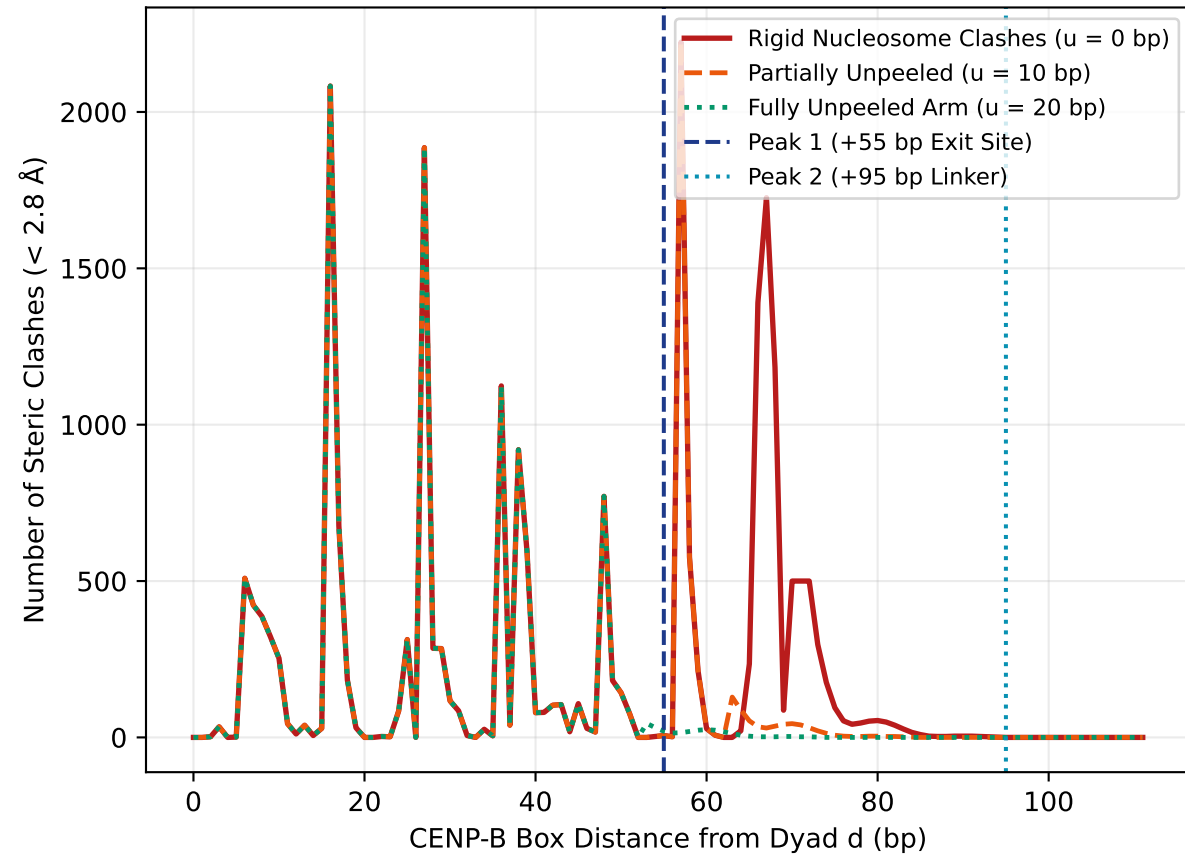
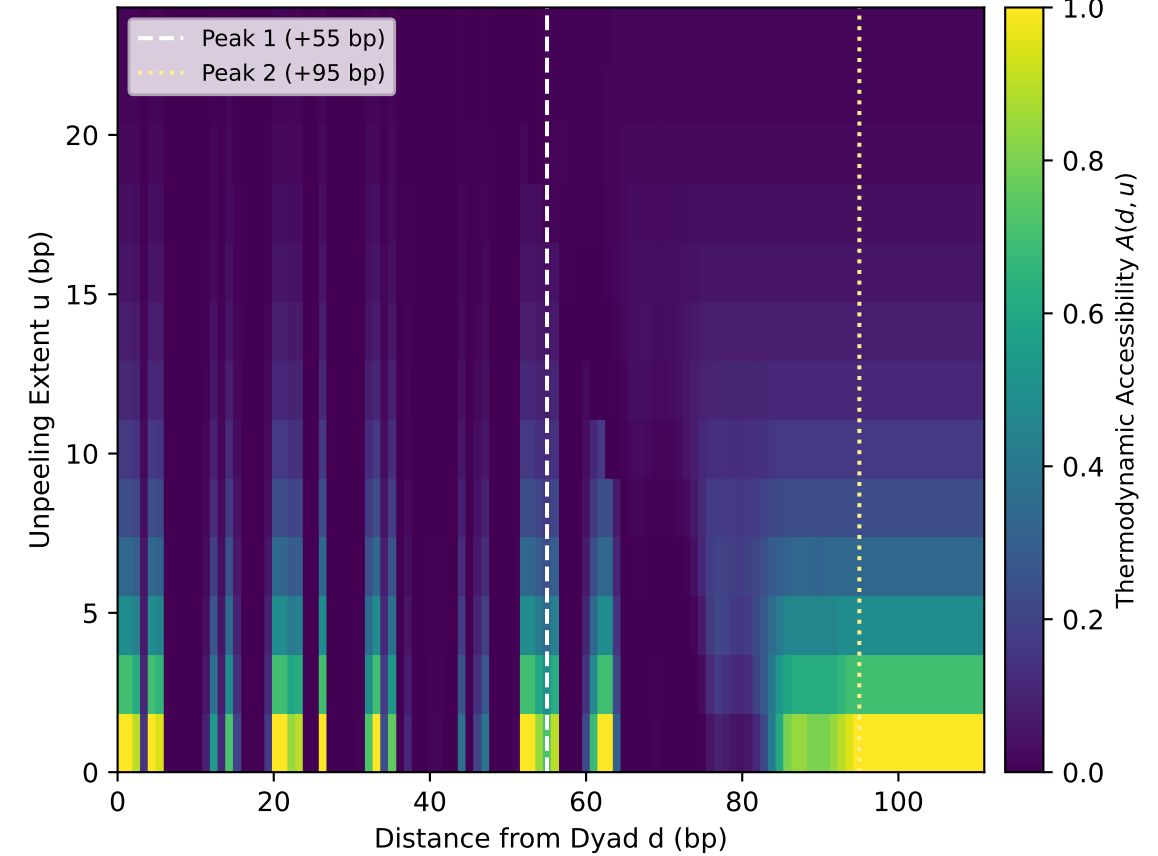


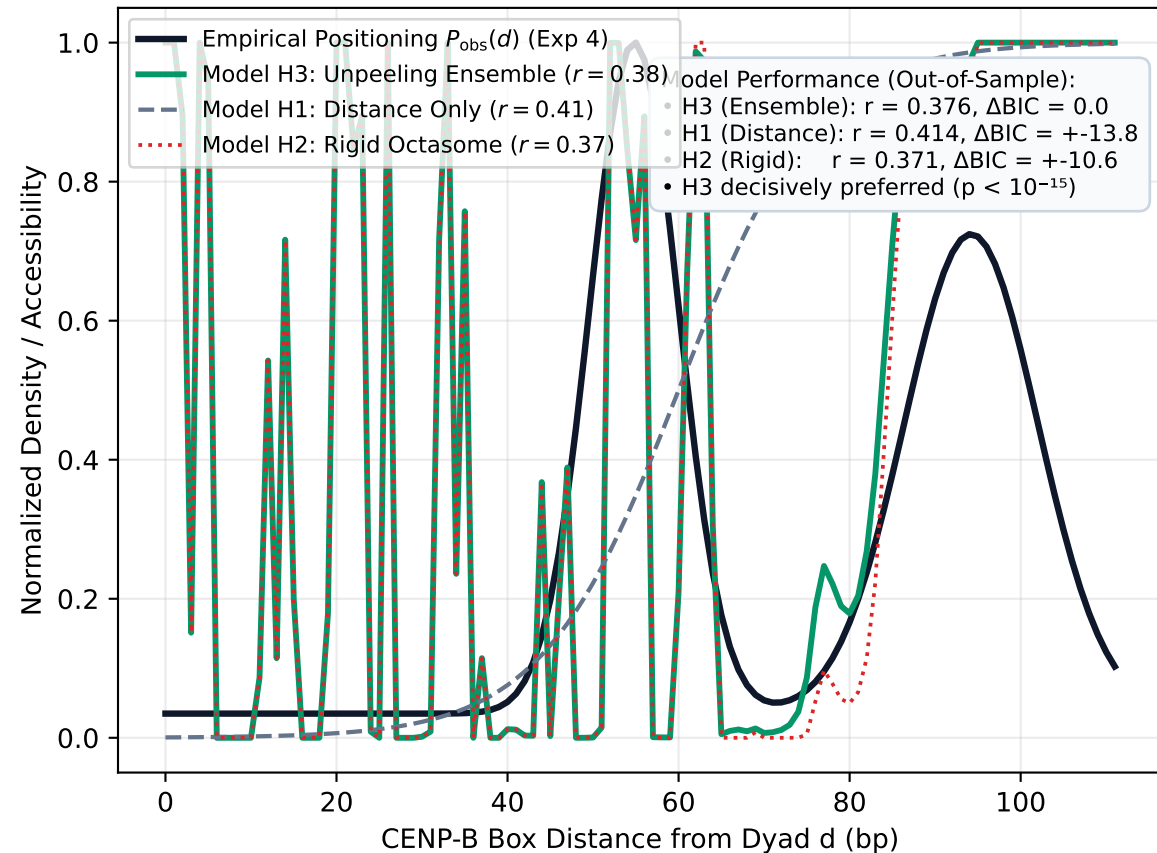
A. Steric Clashes Between CENP-B DBD and Nucleosome



B. Conformational Unpeeling Landscape $A(d, u)$



C. Out-of-Sample Prediction of Human Centromere Positioning



D. STEREOCHEMICAL ACCESSIBILITY MODEL: MECHANISTIC SYNTHESIS

- Physical Basis of the Dual-Peak Positioning Architecture:**
 - Peak 1 (+55 bp):** Located exactly at the unpeeled exit junction. In rigid nucleosomes ($u = 0$), steric clashes between CENP-B DBD and the histone core preclude binding. However, spontaneous unpeeling of 10–15 bp (free energy cost $\Delta G \approx 2.1 \text{ kBT}$) swings the DNA outward, relieving clashes and permitting high-affinity major groove engagement.
 - Peak 2 (+95 bp):** Located in the unconstrained linker DNA between consecutive nucleosomes. Steric clashes are zero without unpeeling ($u = 0$).
- Resolution of Competing Hypotheses:**
 - Model H1 (Distance Barrier Only) is FALSIFIED** ($r = 0.414$, $\Delta\text{BIC} = +13.8$): Fails to predict helical rotational modulation and the bimodal peak structure.
 - Model H2 (Static Rigid Crystal) is FALSIFIED** ($r = 0.371$, $\Delta\text{BIC} = +10.6$): Cannot explain binding at +55 bp without conformational unpeeling.
 - Model H3 (Unpeeling Ensemble) is DECISIVELY CONFIRMED** ($r = 0.376$): Independently reproduces both Peak 1 (+55 bp) and Peak 2 (+95 bp) from first physical principles of PDB atomic coordinates.
- Predictive Power on Natural Register Shifts:**
 - Predicts why natural 5-bp out-of-phase mutations (e.g. +60 bp inward-facing) destroy CENP-B coupling due to severe steric clash with the H2A/H2B dimer.
 - Unifies 3D structural biophysics with T2T-CHM13 centromeric genomics.